



# Storage and File Structure

**Database System Concepts, 6<sup>th</sup> Ed.**

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# Classification of Physical Storage Media

- Speed with which data can be accessed
- Cost per unit of data
- Reliability
  - data loss on power failure or system crash
  - physical failure of the storage device
- Can differentiate storage into:
  - **volatile storage**: loses contents when power is switched off
  - **non-volatile storage**:
    - ▶ Contents persist even when power is switched off.
    - ▶ Includes secondary and tertiary storage, as well as battery-backed up main-memory.



# Physical Storage Media

- **Cache** – fastest and most costly form of storage; volatile; managed by the computer system hardware.
- **Main memory:**
  - fast access (10s to 100s of nanoseconds; 1 nanosecond =  $10^{-9}$  seconds)
  - generally too small (or too expensive) to store the entire database
    - ▶ capacities of up to a few Gigabytes widely used currently
    - ▶ Capacities have gone up and per-byte costs have decreased steadily and rapidly (roughly factor of 2 every 2 to 3 years)
  - **Volatile** — contents of main memory are usually lost if a power failure or system crash occurs.



# Physical Storage Media (Cont.)

## ■ Flash memory

- Data survives power failure
- Data can be written at a location only once, but location can be erased and written to again
  - ▶ Can support only a limited number (10K – 1M) of write/erase cycles.
  - ▶ Erasing of memory has to be done to an entire bank of memory
- Reads are roughly as fast as main memory
- But writes are slow (few microseconds), erase is slower
- Widely used in embedded devices such as digital cameras, phones, and USB keys



# Physical Storage Media (Cont.)

## ■ Magnetic-disk

- Data is stored on spinning disk, and read/written magnetically
- Primary medium for the long-term storage of data; typically stores entire database.
- Data must be moved from disk to main memory for access, and written back for storage
  - ▶ Much slower access than main memory (more on this later)
- **direct-access** – possible to read data on disk in any order, unlike magnetic tape
- Capacities range up to roughly 1.5 TB as of 2009
  - ▶ Much larger capacity and cost/byte than main memory/flash memory
  - ▶ Growing constantly and rapidly with technology improvements (factor of 2 to 3 every 2 years)
- Survives power failures and system crashes
  - ▶ disk failure can destroy data, but is rare



# Physical Storage Media (Cont.)

## ■ Optical storage

- non-volatile, data is read optically from a spinning disk using a laser
- CD-ROM (640 MB) and DVD (4.7 to 17 GB) most popular forms
- Blu-ray disks: 27 GB to 54 GB
- Write-one, read-many (WORM) optical disks used for archival storage (CD-R, DVD-R, DVD+R)
- Multiple write versions also available (CD-RW, DVD-RW, DVD+RW, and DVD-RAM)
- Reads and writes are slower than with magnetic disk
- **Juke-box** systems, with large numbers of removable disks, a few drives, and a mechanism for automatic loading/unloading of disks available for storing large volumes of data



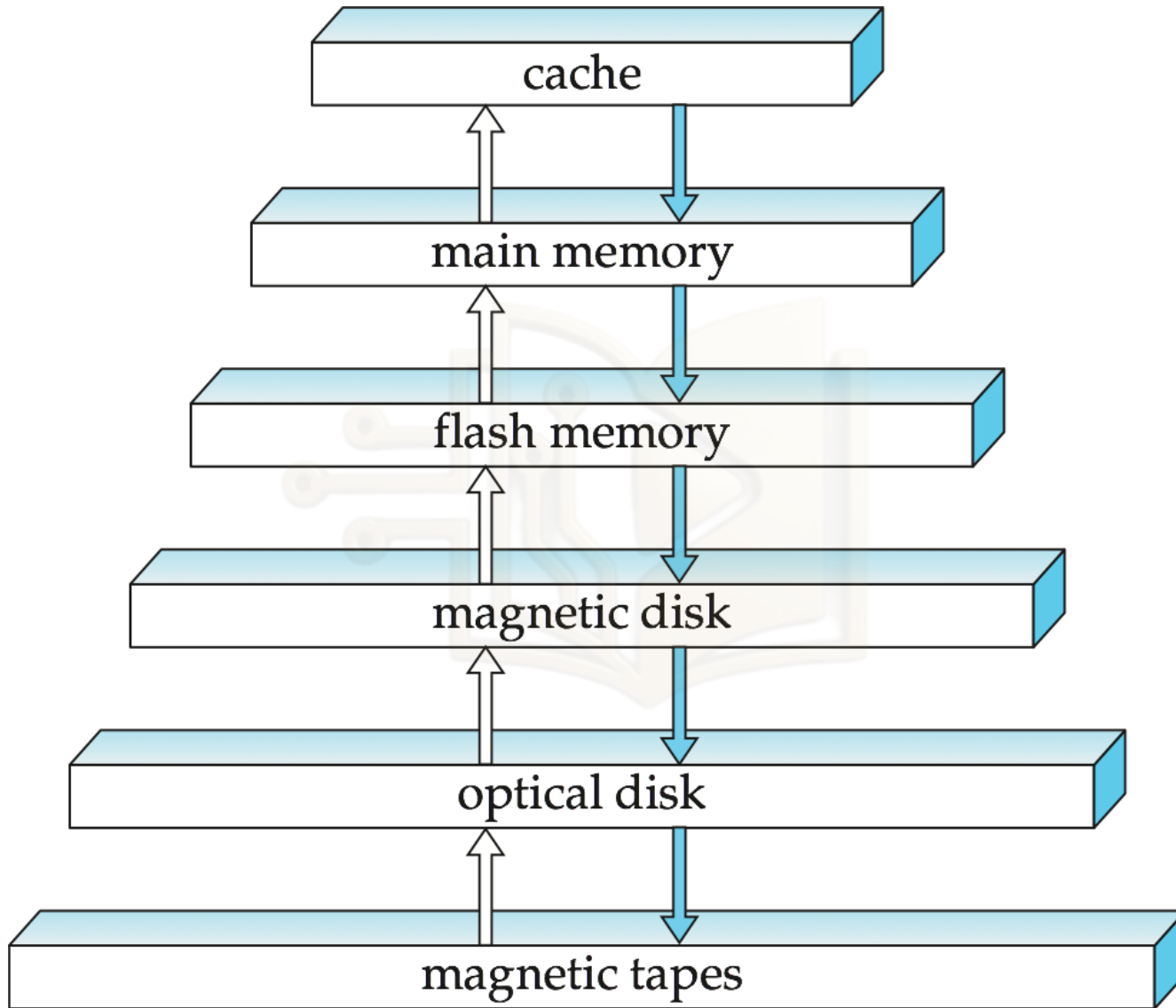
# Physical Storage Media (Cont.)

## ■ Tape storage

- non-volatile, used primarily for backup (to recover from disk failure), and for archival data
- **sequential-access** – much slower than disk
- very high capacity (40 to 300 GB tapes available)
- tape can be removed from drive  $\Rightarrow$  storage costs much cheaper than disk, but drives are expensive
- Tape jukeboxes available for storing massive amounts of data
  - ▶ hundreds of terabytes (1 terabyte =  $10^9$  bytes) to even multiple **petabytes** (1 petabyte =  $10^{12}$  bytes)



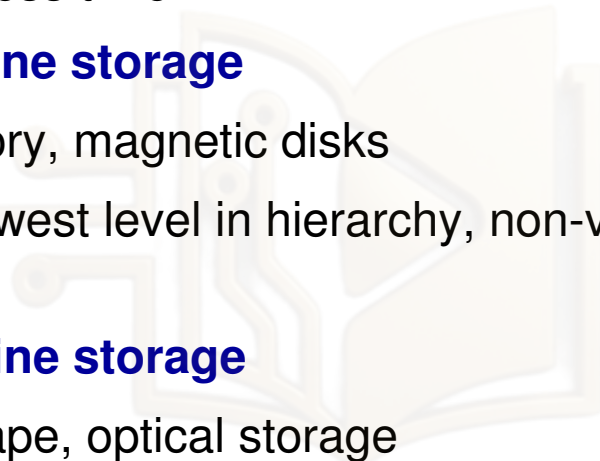
# Storage Hierarchy





# Storage Hierarchy (Cont.)

- **primary storage:** Fastest media but volatile (cache, main memory).
- **secondary storage:** next level in hierarchy, non-volatile, moderately fast access time
  - also called **on-line storage**
  - E.g. flash memory, magnetic disks
- **tertiary storage:** lowest level in hierarchy, non-volatile, slow access time
  - also called **off-line storage**
  - E.g. magnetic tape, optical storage





# Disk Subsystem

- Disks usually connected directly to computer system
- In **Storage Area Networks (SAN)**, a large number of disks are connected by a high-speed network to a number of servers
- In **Network Attached Storage (NAS)** networked storage provides a file system interface using networked file system protocol, instead of providing a disk system interface





# Performance Measures (Cont.)

- **Mean time to failure (MTTF)** – the average time the disk is expected to run continuously without any failure.
  - Typically 3 to 5 years
  - Probability of failure of new disks is quite low, corresponding to a “theoretical MTTF” of 500,000 to 1,200,000 hours for a new disk
    - ▶ E.g., an MTTF of 1,200,000 hours for a new disk means that given 1000 relatively new disks, on an average one will fail every 1200 hours
  - MTTF decreases as disk ages



# RAID

## ■ RAID: Redundant Arrays of Independent Disks

- disk organization techniques that manage a large numbers of disks, providing a view of a single disk of
  - ▶ **high capacity** and **high speed** by using multiple disks in parallel,
  - ▶ **high reliability** by storing data redundantly, so that data can be recovered even if a disk fails
- The chance that some disk out of a set of  $N$  disks will fail is much higher than the chance that a specific single disk will fail.
  - E.g., a system with 100 disks, each with MTTF of 100,000 hours (approx. 11 years), will have a system MTTF of 1000 hours (approx. 41 days)
  - Techniques for using redundancy to avoid data loss are critical with large numbers of disks
- Originally a cost-effective alternative to large, expensive disks
  - I in RAID originally stood for “inexpensive”
  - Today RAIDs are used for their higher reliability and bandwidth.
    - ▶ The “I” is interpreted as independent



# Improvement of Reliability via Redundancy

- **Redundancy** – store extra information that can be used to rebuild information lost in a disk failure
- E.g., **Mirroring** (or **shadowing**)
  - Duplicate every disk. Logical disk consists of two physical disks.
  - Every write is carried out on both disks
    - ▶ Reads can take place from either disk
  - If one disk in a pair fails, data still available in the other
    - ▶ Data loss would occur only if a disk fails, and its mirror disk also fails before the system is repaired
      - Probability of combined event is very small
        - » Except for dependent failure modes such as fire or building collapse or electrical power surges
- **Mean time to data loss** depends on mean time to failure, and **mean time to repair**
  - E.g. MTTF of 100,000 hours, mean time to repair of 10 hours gives mean time to data loss of  $500 \times 10^6$  hours (or 57,000 years) for a mirrored pair of disks (ignoring dependent failure modes)



# Improvement in Performance via Parallelism

- Two main goals of parallelism in a disk system:
  1. Load balance multiple small accesses to increase throughput
  2. Parallelize large accesses to reduce response time.
- Improve transfer rate by striping data across multiple disks.
- **Bit-level striping** – split the bits of each byte across multiple disks
  - In an array of eight disks, write bit  $i$  of each byte to disk  $i$ .
  - Each access can read data at eight times the rate of a single disk.
  - But seek/access time worse than for a single disk
    - ▶ Bit level striping is not used much any more
- **Block-level striping** – with  $n$  disks, block  $i$  of a file goes to disk  $(i \bmod n) + 1$ 
  - Requests for different blocks can run in parallel if the blocks reside on different disks
  - A request for a long sequence of blocks can utilize all disks in parallel



# RAID Levels

- Schemes to provide redundancy at lower cost by using disk striping combined with parity bits
  - Different RAID organizations, or RAID levels, have differing cost, performance and reliability characteristics
- **RAID Level 0: Block striping; non-redundant.**
  - Used in high-performance applications where data loss is not critical.
- **RAID Level 1: Mirrored disks** with block striping
  - Offers best write performance.
  - Popular for applications such as storing log files in a database system.



(a) RAID 0: nonredundant striping



(b) RAID 1: mirrored disks



# RAID Levels (Cont.)

- **RAID Level 2:** Memory-Style Error-Correcting-Codes (ECC) with bit striping.
- **RAID Level 3:** Bit-Interleaved Parity
  - a single parity bit is enough for error correction, not just detection, since we know which disk has failed
    - ▶ When writing data, corresponding parity bits must also be computed and written to a parity bit disk
    - ▶ To recover data in a damaged disk, compute XOR of bits from other disks (including parity bit disk)



(c) RAID 2: memory-style error-correcting codes



(d) RAID 3: bit-interleaved parity



# RAID Levels (Cont.)

## ■ RAID Level 3 (Cont.)

- Faster data transfer than with a single disk, but fewer I/Os per second since every disk has to participate in every I/O.
- Subsumes Level 2 (provides all its benefits, at lower cost).

## ■ RAID Level 4: Block-Interleaved Parity; uses block-level striping, and keeps a parity block on a separate disk for corresponding blocks from $N$ other disks.

- When writing data block, corresponding block of parity bits must also be computed and written to parity disk
- To find value of a damaged block, compute XOR of bits from corresponding blocks (including parity block) from other disks.



(e) RAID 4: block-interleaved parity



# RAID Levels (Cont.)

## ■ RAID Level 4 (Cont.)

- Provides higher I/O rates for independent block reads than Level 3
  - ▶ block read goes to a single disk, so blocks stored on different disks can be read in parallel
- Provides high transfer rates for reads of multiple blocks than no-striping
- Before writing a block, parity data must be computed
  - ▶ Can be done by using old parity block, old value of current block and new value of current block (2 block reads + 2 block writes)
  - ▶ Or by recomputing the parity value using the new values of blocks corresponding to the parity block
    - More efficient for writing large amounts of data sequentially
- Parity block becomes a bottleneck for independent block writes since every block write also writes to parity disk



# RAID Levels (Cont.)

- **RAID Level 5: Block-Interleaved Distributed Parity**; partitions data and parity among all  $N + 1$  disks, rather than storing data in  $N$  disks and parity in 1 disk.
  - E.g., with 5 disks, parity block for  $n$ th set of blocks is stored on disk  $(n \bmod 5) + 1$ , with the data blocks stored on the other 4 disks.



(f) RAID 5: block-interleaved distributed parity

|    |    |    |    |    |
|----|----|----|----|----|
| P0 | 0  | 1  | 2  | 3  |
| 4  | P1 | 5  | 6  | 7  |
| 8  | 9  | P2 | 10 | 11 |
| 12 | 13 | 14 | P3 | 15 |
| 16 | 17 | 18 | 19 | P4 |



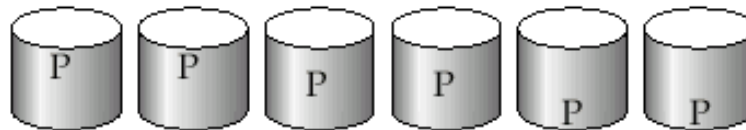
# RAID Levels (Cont.)

## ■ RAID Level 5 (Cont.)

- Higher I/O rates than Level 4.
  - ▶ Block writes occur in parallel if the blocks and their parity blocks are on different disks.
- Subsumes Level 4: provides same benefits, but avoids bottleneck of parity disk.

## ■ RAID Level 6: P+Q Redundancy scheme; similar to Level 5, but stores extra redundant information to guard against multiple disk failures.

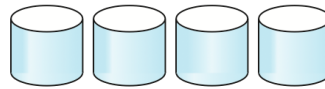
- Better reliability than Level 5 at a higher cost; not used as widely.



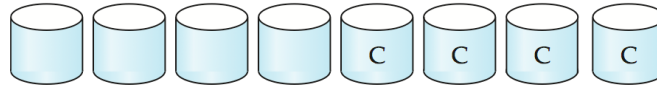
(g) RAID 6: P + Q redundancy



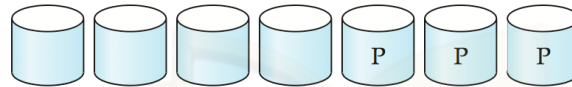
# Figure 10.03



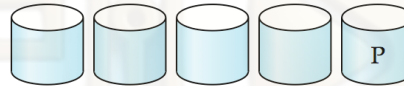
(a) RAID 0: nonredundant striping



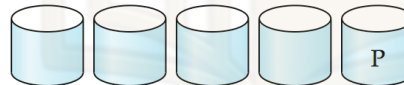
(b) RAID 1: mirrored disks



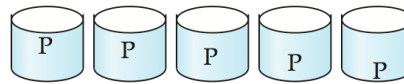
(c) RAID 2: memory-style error-correcting codes



(d) RAID 3: bit-interleaved parity



(e) RAID 4: block-interleaved parity



(f) RAID 5: block-interleaved distributed parity



(g) RAID 6: P + Q redundancy



# Choice of RAID Level

- Factors in choosing RAID level
  - Monetary cost
  - Performance: Number of I/O operations per second, and bandwidth during normal operation
  - Performance during failure
  - Performance during rebuild of failed disk
    - ▶ Including time taken to rebuild failed disk
- RAID 0 is used only when data safety is not important
  - E.g. data can be recovered quickly from other sources
- Level 2 and 4 never used since they are subsumed by 3 and 5
- Level 3 is not used anymore since bit-striping forces single block reads to access all disks, wasting disk arm movement, which block striping (level 5) avoids
- Level 6 is rarely used since levels 1 and 5 offer adequate safety for most applications



# Choice of RAID Level (Cont.)

- Level 1 provides much better write performance than level 5
  - Level 5 requires at least 2 block reads and 2 block writes to write a single block, whereas Level 1 only requires 2 block writes
  - Level 1 preferred for high update environments such as log disks
- Level 1 had higher storage cost than level 5
  - disk drive capacities increasing rapidly (50%/year) whereas disk access times have decreased much less (x 3 in 10 years)
  - I/O requirements have increased greatly, e.g. for Web servers
  - When enough disks have been bought to satisfy required rate of I/O, they often have spare storage capacity
    - ▶ so there is often no extra monetary cost for Level 1!
- Level 5 is preferred for applications with low update rate, and large amounts of data
- Level 1 is preferred for all other applications



# Hardware Issues

- **Software RAID:** RAID implementations done entirely in software, with no special hardware support
- **Hardware RAID:** RAID implementations with special hardware
  - Use non-volatile RAM to record writes that are being executed
  - Beware: power failure during write can result in corrupted disk
    - ▶ E.g. failure after writing one block but before writing the second in a mirrored system
    - ▶ Such corrupted data must be detected when power is restored
      - Recovery from corruption is similar to recovery from failed disk
      - NV-RAM helps to efficiently detected potentially corrupted blocks
        - » Otherwise all blocks of disk must be read and compared with mirror/parity block



# Hardware Issues (Cont.)

- **Latent failures:** data successfully written earlier gets damaged
  - can result in data loss even if only one disk fails
- **Data scrubbing:**
  - continually scan for latent failures, and recover from copy/parity
- **Hot swapping:** replacement of disk while system is running, without power down
  - Supported by some hardware RAID systems,
  - reduces time to recovery, and improves availability greatly
- Many systems maintain **spare disks** which are kept online, and used as replacements for failed disks immediately on detection of failure
  - Reduces time to recovery greatly
- Many hardware RAID systems ensure that a single point of failure will not stop the functioning of the system by using
  - Redundant power supplies with battery backup
  - Multiple controllers and multiple interconnections to guard against controller/interconnection failures